

Craftax runs compared: achievements and what unlocks them

10 finished runs, 1.1M env steps each, size50m, one training seed per run. Scored with tools/death_eval.py on the same fixed evaluation worlds. Updated 2026-10-03.

Every number is the mean episode-final achievement count over the evaluation episodes, and each unlock rate is the share of episodes in which that achievement fired. Every run was scored on the same fixed evaluation worlds; re-scoring two old checkpoints with today's code reproduced them exactly.

Experiment B sets the record. Masking impossible actions during training (B2) reaches 7.60 achievements, against 5.87 for the same model without it; giving the agent the validity flags as an input (B1) reaches 6.90. The furnace, never placed in any earlier run despite hundreds of chances, is placed in 40% of B2's episodes; it also places stone (40%), mines coal (20%) and, for the first time, makes and places a torch (3%).

The non-cheating model (Honest map) matches the privileged control (5.87 vs 5.93), so dropping privileged supervision cost nothing measurable.

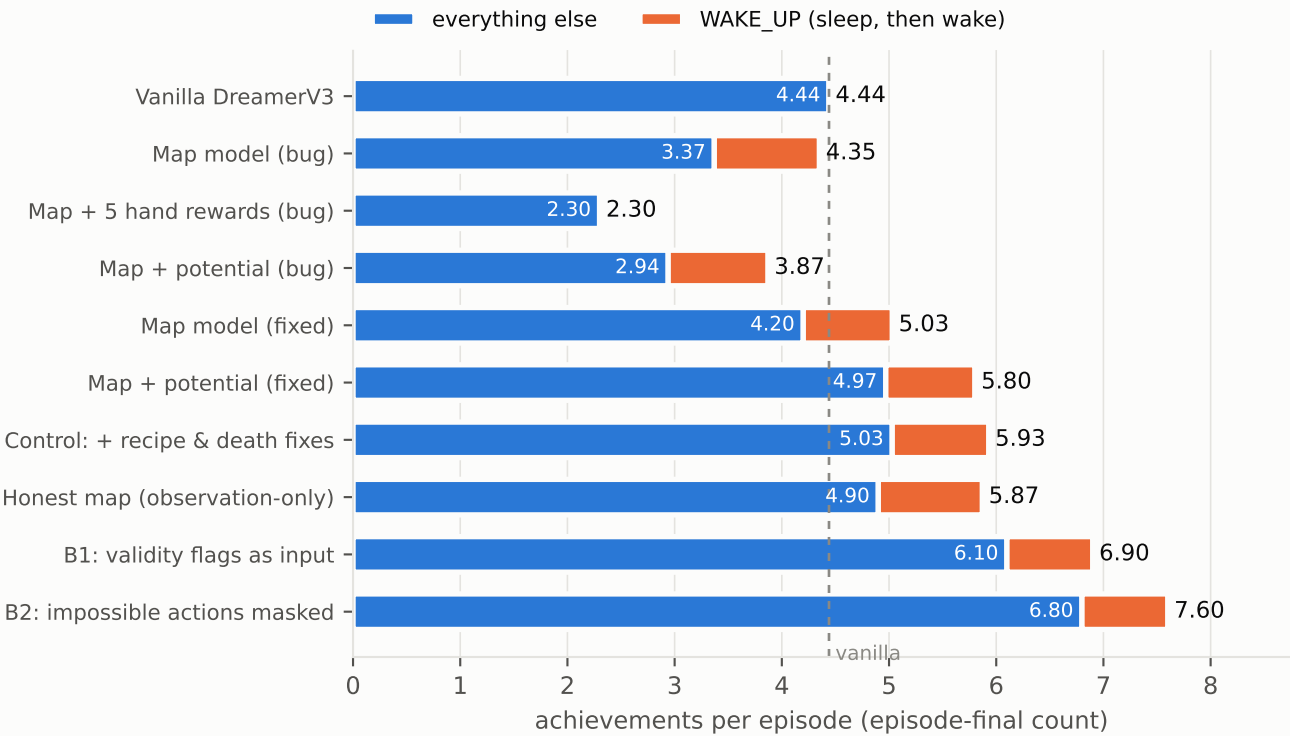
Every map run except the hand-written-rewards one unlocks WAKE_UP (sleep, then wake) in 80-98% of episodes; vanilla never does. Without it the map model alone scores 4.20 against vanilla's 4.44; the gains that remain come from the potential and from experiment B (B2 6.80, B1 6.10 without WAKE_UP). Survival is still unsolved: lifespans 243-311 steps against 261-284 for a random policy. In normalized return the best run is about 3.0% of the 226-point maximum, against roughly 7% reported for the state of the art.

run	eps	ach.	w/o wake	life	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	25	4.44	4.44	262	64%	24%	0%	12%	80%
Map model (bug)	60	4.35	3.37	294	35%	0%	0%	0%	43%
Map + 5 hand rewards (bug)	50	2.30	2.30	263	0%	0%	0%	0%	44%
Map + potential (bug)	30	3.87	2.94	277	13%	0%	0%	0%	37%
Map model (fixed)	30	5.03	4.20	264	63%	7%	0%	0%	70%
Map + potential (fixed)	30	5.80	4.97	278	90%	30%	10%	20%	60%
Control: + recipe & death fixes	30	5.93	5.03	268	83%	0%	60%	0%	67%
Honest map (observation-only)	30	5.87	4.90	243	87%	20%	20%	7%	83%
B1: validity flags as input	30	6.90	6.10	311	87%	50%	30%	33%	80%
B2: impossible actions masked	30	7.60	6.80	246	87%	53%	20%	43%	70%

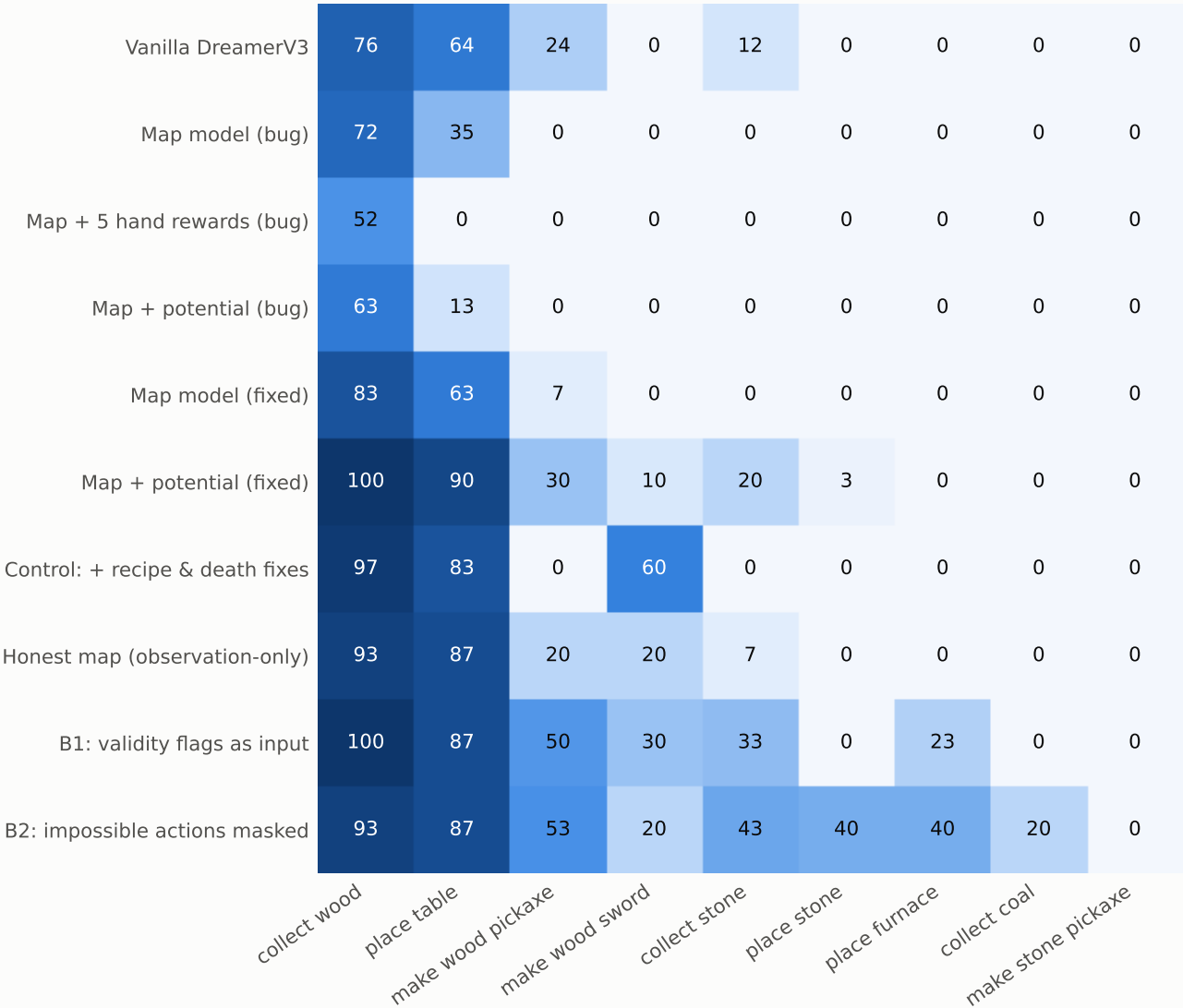
eps = evaluation episodes. ach. = mean episode-final achievements; w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. Rates are % of episodes. 'bug' = imag_shift True, which corrupted the policy gradient (fixed 2026-09-20). map_survive was re-evaluated on 2026-09-28 and reproduced its original 2.30. One training seed per run, so the run-to-run spread is unknown. 'Map + potential (fixed)' is the old 5.80 record; 'Control' adds the recipe and death fixes with the privileged map; 'Honest map' has the same fixes with observation-only targets. 'B1' and 'B2' are experiment B on the honest-map configuration: validity flags as an encoder input, and impossible actions masked in training.

Sleep inflates the map runs; experiment B is real progress

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs



Unlock rate (% of episodes), early tech tree in dependency order



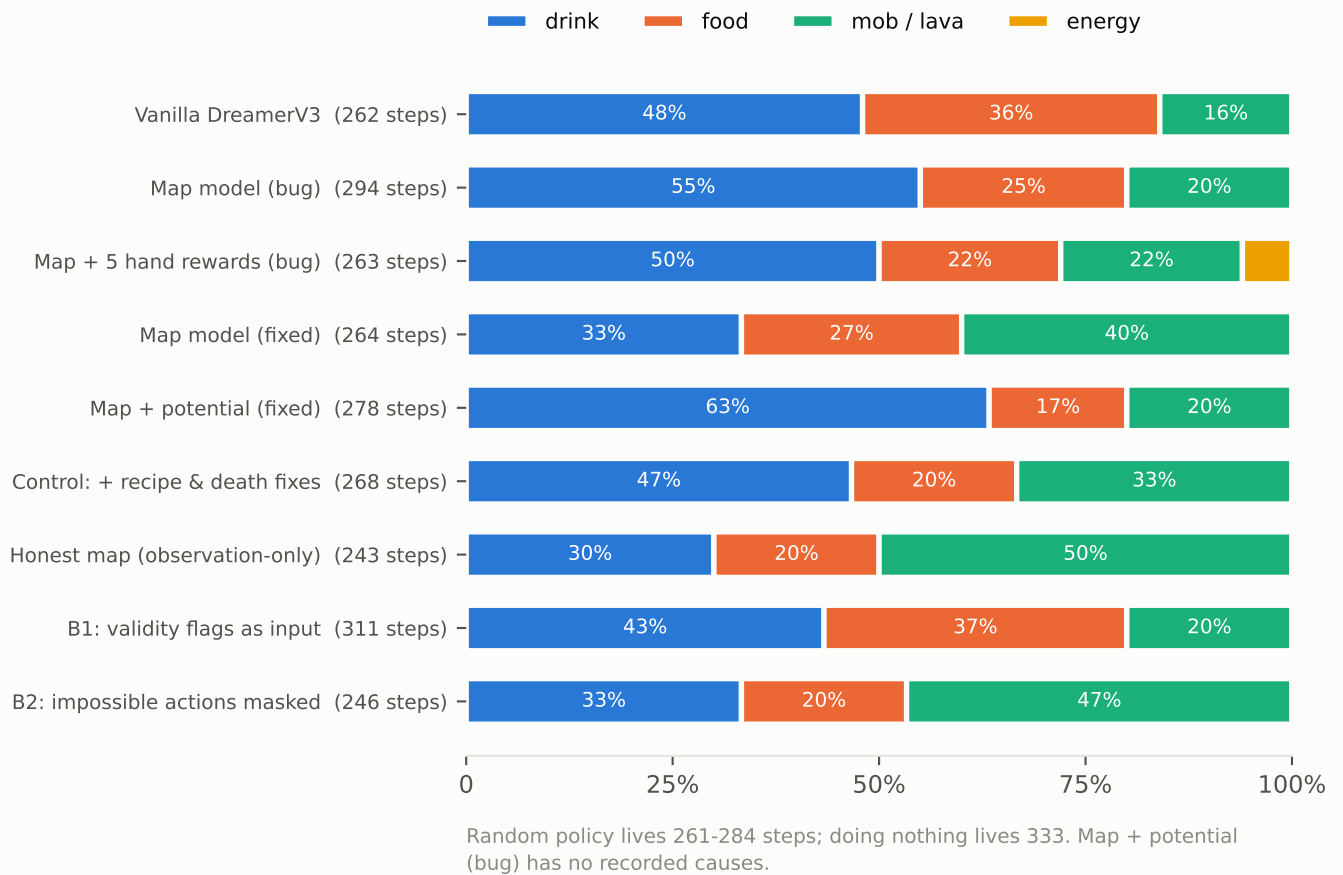
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 50 of 67 were never unlocked by any run.

[illegible]

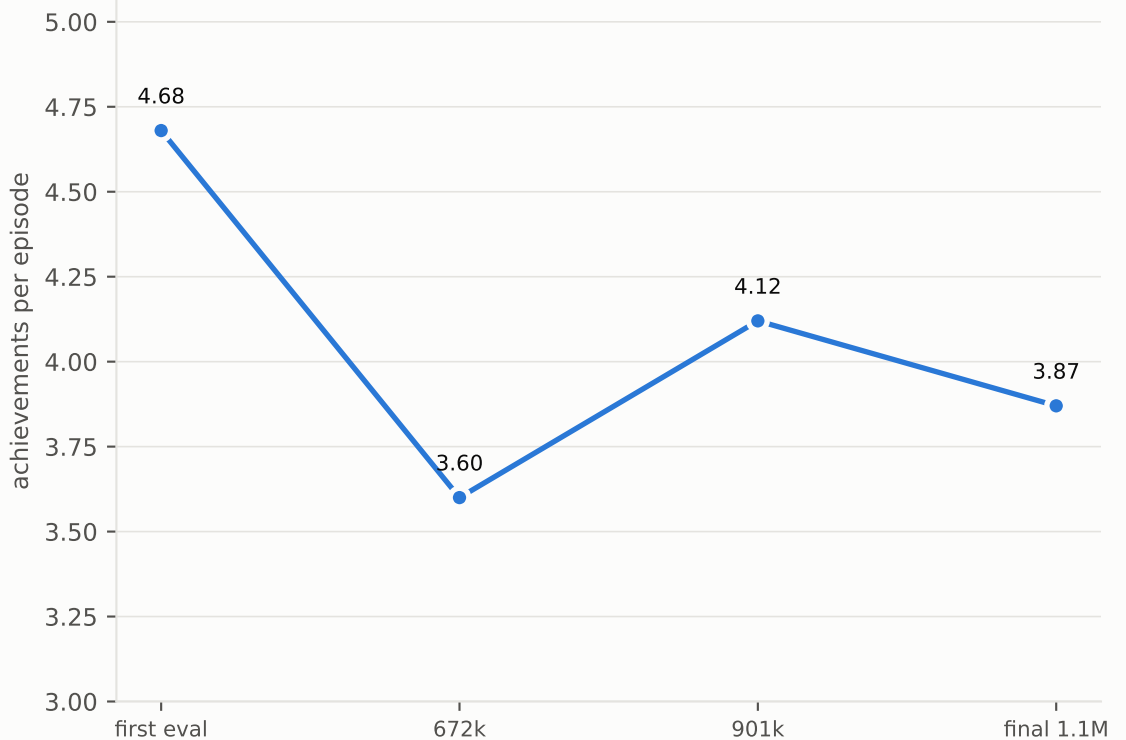
Survival never moved

Cause of death (share of evaluation episodes) and lifespan; and how much one run moves between its own checkpoints



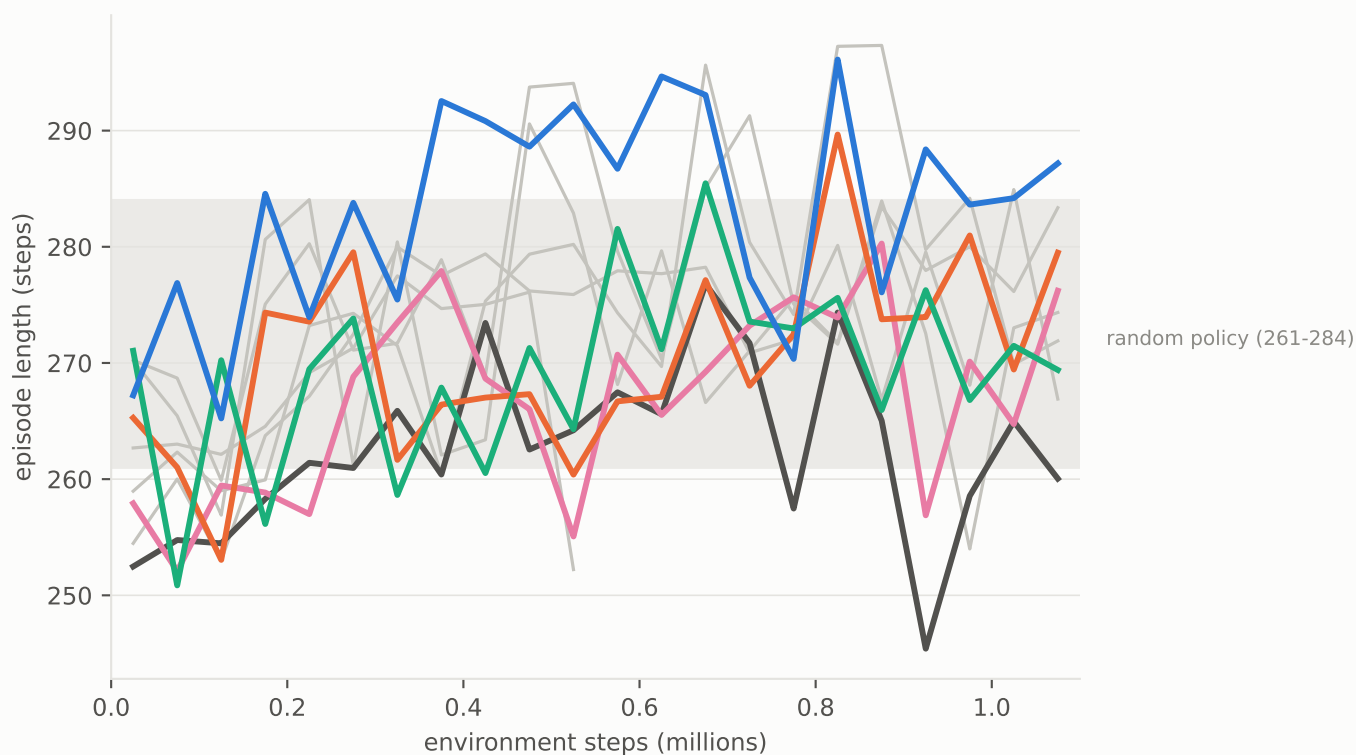
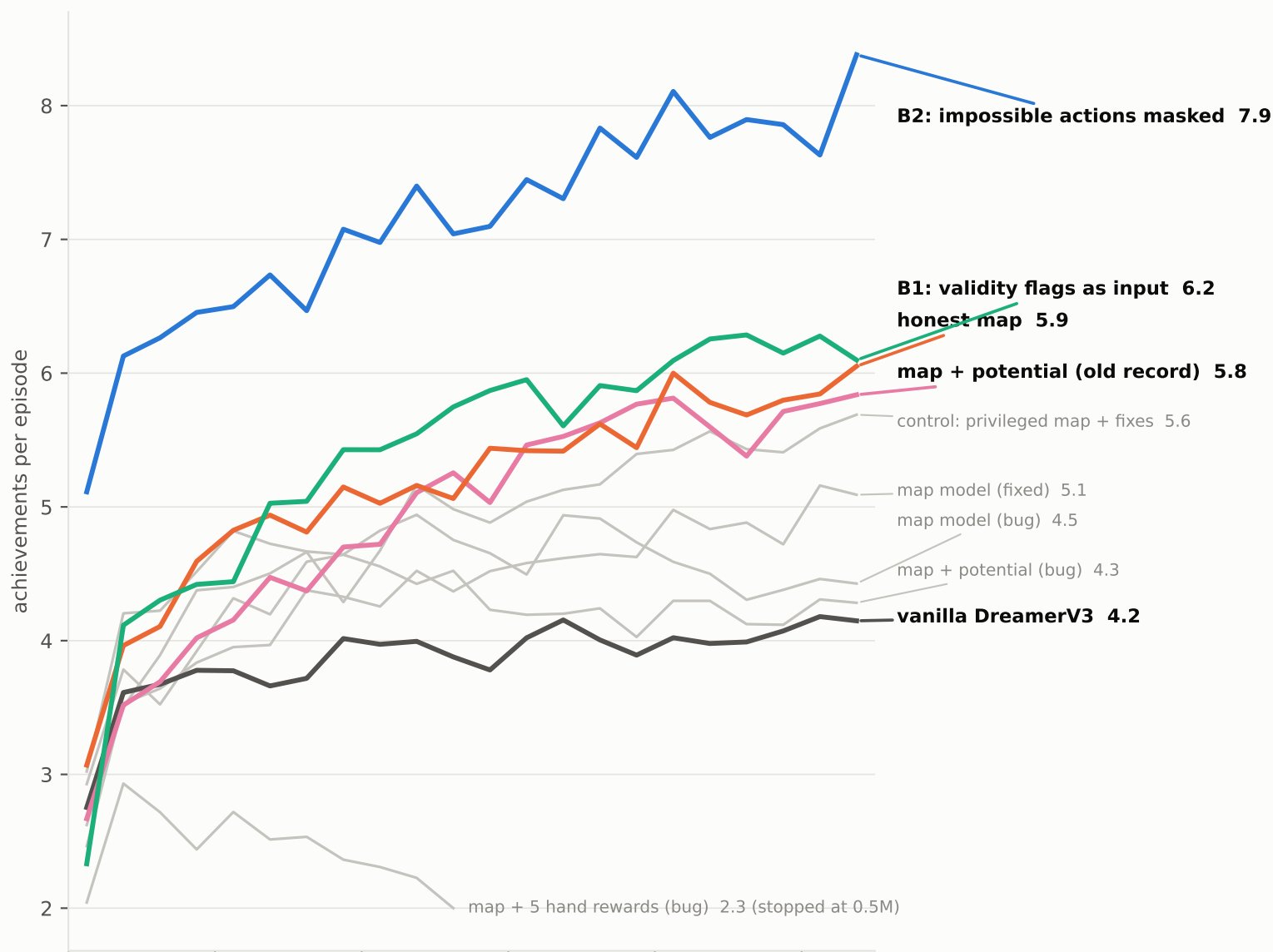
One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



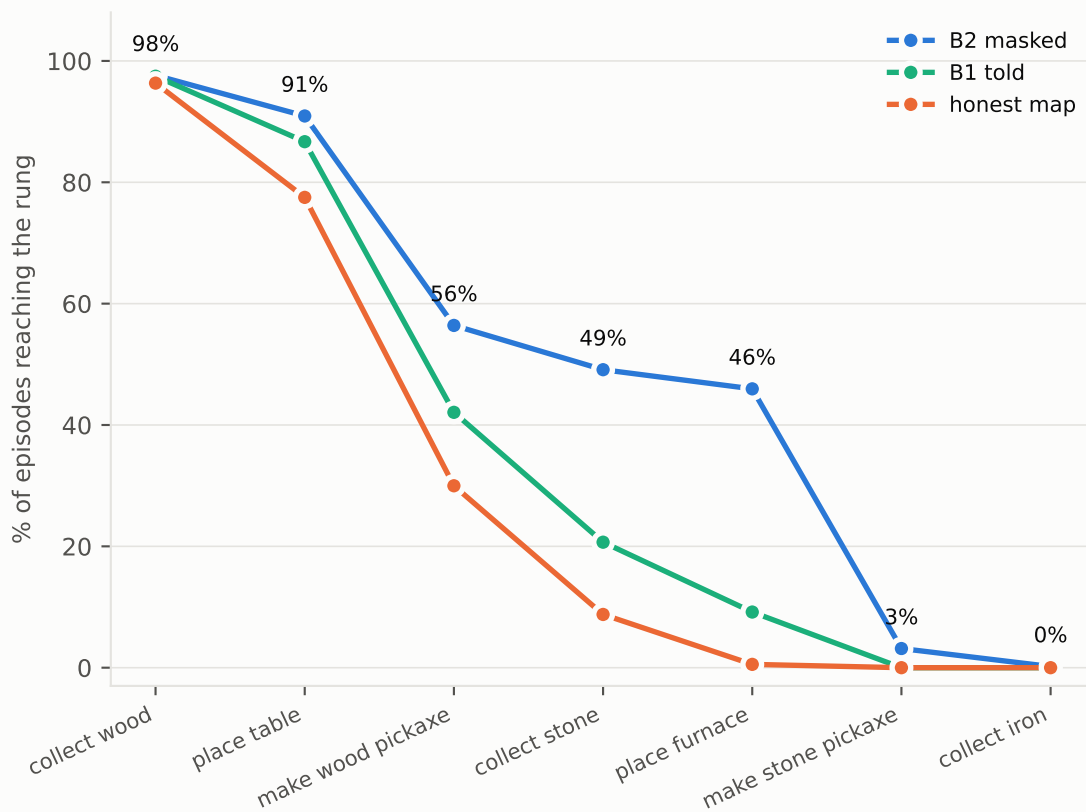
Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps; one seed per run.



Where B2 falls short

Last 150k training steps of each run (~500 episodes), decoded from replay by tools/diagnose_run.py



1. The stone pickaxe is the wall. B2 reaches the furnace in 46% of episodes but makes the stone pickaxe in 3%. Of the 249 episodes that collected stone, 73% held wood and stone at the same time, but only 22% ever stood next to a table while holding both, and 6% made the pickaxe. Two gaps: it rarely brings the ingredients back to a table (logistics), and when it does the chance lasts one step -- the single log usually goes into the wood sword or it walks off (measured in docs/b2-casebook.pdf).
2. Table to wood pickaxe leaks. 38% of the episodes that place a table never make the wood pickaxe. At a table holding wood it crafts every time, but every one of these episodes placed the table with its last two logs, and later logs go into another table rather than back to the first (docs/b2-casebook.pdf).
3. Survival. Training-window lifespan 285 (median 275) against 261-284 for a random policy. Thirst is still the first meter to run out in 45% of deaths and hunger in 24%: Crafttax pays for the first drink only, so topping up is never rewarded. A hostile mob is adjacent at 61% of deaths; it dies in lava in 4% (the others: under 1%), the price of exploring further. Deaths in its sleep are rare (5%).
4. Wasted presses. 43% of presses still do nothing (experiment A): DO with nothing in front (26%), NOOP and walking into walls. Masking only removes the special actions; these basic ones are never masked.

What the results say, and what would settle the rest

1. Masking impossible actions is the largest gain so far

B2 7.60 vs honest map 5.87, same model and reward. Experiment A on its final checkpoint shows why: buttons that sat at exactly 0% in the states where they would work now get pressed there (furnace 3.2% over 599 states, placing stone 2.4%, placing a table 75%, the wood pickaxe 100%), and every no-effect press is now a basic action (DO, NOOP, a blocked move). Once a useless press can no longer teach the agent that a button is bad, it presses that button when it works.

2. Telling helps, masking helps more

B1 (flags as input) reaches 6.90: both wood crafts survive in one run for the first time and the furnace comes off zero, but placing stone stays at 0.0% across 452 chances and about 6% of presses still go to impossible buttons. The information helps; removing the suppressing gradient helps more. Masking changes the action space relative to published Craftax baselines, so it is reported as a separate arm.

3. The next wall is sequencing, not suppression

In B2 the stone pickaxe was craftable in only 5 of 5,167 evaluation steps: the agent mines stone but rarely returns to a table holding both wood and stone. That gates iron and every 3-5-point achievement after it. It is a planning problem, so the next experiment should target multi-step sequences (experiment D, return-then-explore, or a sub-goal signal), not the action distribution.

4. Survival is still unsolved

Lifespans stay at or below the random band, except B1 (311, mean pulled up by a few long episodes; median 265). B2 dies earliest (246) and half of its deaths are mobs or lava: the more it does, the more it is exposed.

5. What to run next

Second seeds of B2 and the honest map, since every comparison here is single-seed. Then a sequencing experiment for the stone pickaxe. Explain the WAKE_UP gap before claiming the map model helps.